

Level and Shape: Learning Adjacent-Lanthanide Selectivity from the Solvent-Extraction Record

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Abstract

Separating neighbouring lanthanides is the most demanding step in rare-earth processing. Adjacent ions differ in radius by about 0.013 Å, and a ligand that prefers one over the other is still found by trial. We ask how well that preference can be predicted for ligands a model has never seen, using nothing but the published solvent-extraction literature. From the SAFE database we assembled 4,746 distribution ratios for 162 extractants across the lanthanide series and trained gradient-boosting models to predict the separation factor of each adjacent pair measured under the same conditions. With whole extractants held out, predicted and measured log separation factors correlate at $r = 0.59$ over 905 pairs ($R^2 = 0.33$). The improvement over a model trained on the distribution ratios directly ($R^2 = 0.27$) comes from one change: a ligand’s overall extraction strength and the differences between metals are predicted by two separate models, so the second is judged only on what tells the metals apart. In practice, half of the model’s top-quartile suggestions for a given pair are genuinely top-quartile (a quarter by chance), separation factors above two are picked out with an AUC of 0.81, and the uncertainty intervals hold their nominal coverage. A graph neural network on computed structures of the metal–ligand complexes adds only 0.015 to the score, and a benchmark of three semiempirical methods on 71 ligands shows why: GFN2-xTB reproduces 40 % of the lanthanide contraction and g-xTB 108 %, but under either the structures carry little about the metal beyond its size, which the model already has. The score stops rising at about forty-five training extractants, so further gains will come from information that separates ligands within a chemical family, not from more of the same data.

1 Introduction

Trivalent lanthanides are chemically alike. The ionic radius shrinks by roughly 0.013 Å per element along the series [1], and that contraction is nearly all a ligand has to work with when asked to prefer neodymium over praseodymium. Industrial separation therefore runs through dozens of extraction stages, and the search for better extractants is still largely done by hand [2].

Decades of that hand work are written down. The SAFE database [3] collects distribution ratios from the extraction literature, and many ligands in it have been measured across most of the series under one set of conditions. Two distribution ratios from the same experiment give the separation factor of that pair,

with the noise common to both cancelled. This paper asks how much adjacent-lanthanide selectivity can be learned from that record for a ligand the model has never seen, and what a 3D structure of the complex adds. The answer has four parts: a scoring construction that isolates selectivity from extraction strength (Section 2), the level/shape decomposition (Section 3), what 3D structure adds and why so little (Sections 4 and 5), and how much data the problem needs.

2 Data and target

From SAFE we keep lanthanide(III) rows with a resolvable extractant structure, canonicalise the SMILES, remove duplicates and detection-limit placeholders, and retain the 4,746 measurements of $\log D$ on 162 extractants for which a complex geometry exists, so that every model is compared on the same rows; each row carries the extractant, the metal and 64 condition columns. The 956 complexes were built with Architector [4] (extractant plus nitrate or water to coordination number 8–10) and optimised with GFN2-xTB [5].

A separation factor is only defined between measurements made under the same conditions, so rows are grouped into blocks (one extractant, one bin of each of ten condition descriptors), replicate measurements of a metal within a block are averaged into a cell, and each pair of cells whose lanthanides are neighbours in the series gives one target value,

$$\log \text{SF}_{i,i+1} = \log D_{\text{Ln}_i} - \log D_{\text{Ln}_{i+1}}. \quad (1)$$

Models are trained on $\log D$ row by row and evaluated after this construction is applied to their out-of-fold predictions; the score R_{adj}^2 is the coefficient of determination of predicted against measured $\log \text{SF}$ over all adjacent pairs. The differencing is what makes the problem tractable: raw $\log D$ scatters by 0.7–0.9 within a (block, metal) group, adjacent separations reproduce to about 0.16.

Every number below is out-of-fold under five-fold cross-validation grouped by extractant, repeated three times. The development population has 905 adjacent pairs on 76 extractants; a second set of 444 pairs, from rows outside it, was set aside before any 3D model was built and used only for the final test; their union has 1,349 pairs.

3 Model

The tabular features are RDKit descriptors and 2048-bit Morgan fingerprints of the extractant [6], the metal’s atomic number, position in the series and Shannon radius, and the 64 condition columns, 746 in all. The learner is CatBoost [7] with a quantile loss ($\alpha = 0.6$), depth 9 and feature subsampling 0.3, chosen on the adjacent-pair score. Trained on $\log D$ directly it reaches $R_{\text{adj}}^2 = +0.268$.

The raw target mixes two questions, how strongly a ligand extracts under given conditions and how that changes from one metal to the next, and only the second is scored. So two models answer them. A level model f_L is trained on $\log D$ and used only through its mean over the test block, the anchor a_b ; a shape model f_S is trained on the block-centred target $\log D_i - \bar{y}_{b(i)}$ and is itself centred within the test block,

$$\hat{y}_i = a_{b(i)} + \left(f_S(x_i) - \text{mean}_{j \in b(i)} f_S(x_j) \right). \quad (2)$$

Nothing else changes; the shape model can no longer trade accuracy on metal differences for accuracy on the level, and the score rises to +0.318 over eight seeds.

For the 3D channel, heavy atoms of the complex are nodes, atoms within 4 Å are connected, and three continuous-filter layers of width 96 [8] feed a readout with a metal-centred term and a radial shell histogram, since plain mean pooling averages away shifts of a few hundredths of an Ångström; the embedding is concatenated with the tabular features, and training uses a Huber loss on $\log D$ plus a contrast term on the within-block pairs of each batch (Appendix A). On its own the network scores +0.266, no better than the boosting baseline. Its block-centred prediction s_{3D} enters Eq. (2) as a second shape source, $\hat{y} = a + (1 - w) s_{\text{tab}} + w s_{3D}$, with $w = 0.35$ from a nested criterion; the score becomes +0.326. A simplicial variant with triangle message passing makes the same predictions ($r = 0.963$), so we report the distance graph.

4 Results

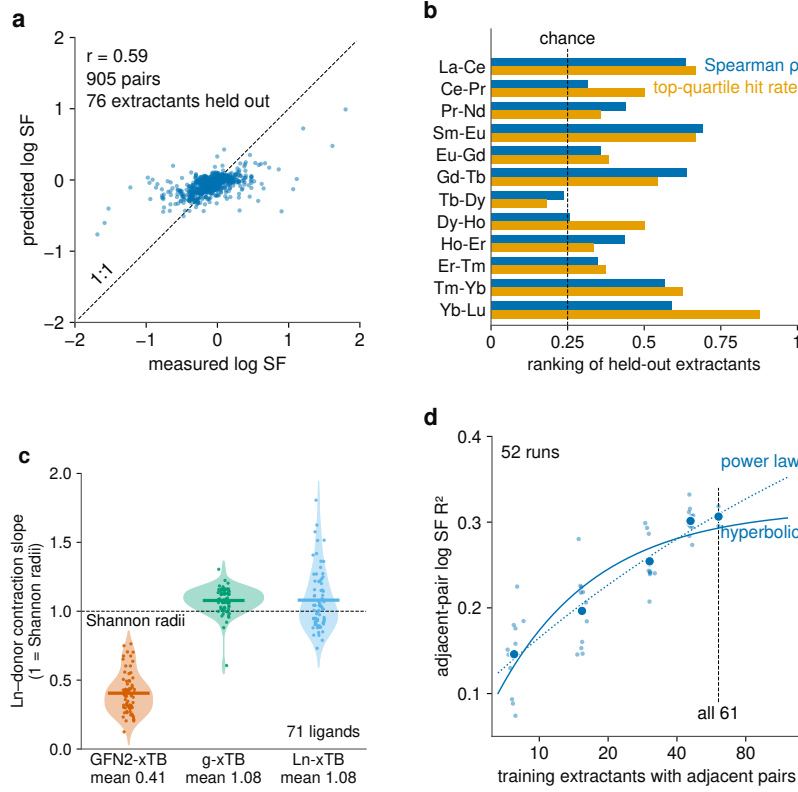


Figure 1: (a) Predicted against measured adjacent-pair log SF for the 905 development pairs, extractants held out; the dashed line is the identity. (b) Per adjacent pair: Spearman ρ between predicted and measured extractant rankings, and the top-quartile hit rate against chance. (c) Slope of the Ln-donor distance against the Shannon radius (the compliance c_L) for the 71 ligands under three Hamiltonians: points are ligands, shaded shapes their kernel density, bars the mean; 1 is exact agreement with the radii. (d) Learning curve: small points are the 52 individual runs (four CV seeds, three nested draws) at their own training size, large markers the mean per fraction; solid and dotted lines the hyperbolic and power-law extrapolations; dashed line the full training set.

The correlation is real and it is ligand-specific. Predicted and measured separation factors correlate at $r = 0.59$ on held-out extractants (Figure 1a); a cluster bootstrap over extractants puts R^2_{adj} between +0.21 and +0.38, and shuffling predictions within each pair position, which keeps the series shape all ligands share, gives -0.05 . With position means and extractant means both removed (10 and 13 % of the variance), the model still scores +0.23 on the ligand-by-position interaction, which is the selectivity itself.

What a screening user gets. Ten percent of the pairs carry 71 % of the variance, so R^2_{adj} mostly describes large separations; the questions a user asks are better answered directly (Figure 1b). Ranking held-out extractants for a given adjacent pair gives a mean Spearman correlation of 0.46 over the twelve positions, and the predicted top quartile contains a measured top-quartile ligand 50 % of the time against 25 % by chance. As a detector of separations with $\text{SF} \geq 2$ (base rate 0.20) the same predictions give AUC 0.81. Split-conformal intervals [9], calibrated leave-one-extractant-out, cover 79 % and 90 % at nominal 80 and 90.

Table 1: Adjacent-pair logSF R^2 , extractants held out (development 905 pairs; expanded 1,220). Pair identity alone scores +0.066.

Model	Development	Expanded
CatBoost on log D	+0.268	–
3D encoder, standalone	+0.266	–
Level + shape, Eq. (2)	+0.318	+0.280
Level + shape + 3D shape, $w = 0.35$	+0.326	+0.288

The 3D gain holds on held-out pairs. The blend improves on the level/shape model by less than the seed-to-seed spread of one encoder run (sd 0.0285), so we fixed $w = 0.35$ in advance and scored the 444 held-out pairs once: the blend scored +0.1207 against +0.1051 for the tabular model, and the same contrast is +0.0146 on the development pairs and +0.0149 on the union of 1,349, all against a seed-matched anchored baseline. Geometry helps when it can only move the shape; as 84 derived columns in a flat model it takes the score from +0.278 to −0.022.

5 What 3D structure can add, and how to check before training

Eight 3D encoders trained on these complexes, spanning two architectures, three neighbourhood cutoffs and several losses, are almost interchangeable: disjoint eight-seed halves of one configuration agree at $r = 0.990$, ensembles of two different configurations at 0.986, and the effective rank of the eight prediction vectors is 1.05, whereas a tabular arm of matched strength agrees with a 3D arm at only 0.893.

The reason is in the parameter file: GFN2-xTB fits Ce and Lu and interpolates every lanthanide parameter linearly in Z between them (residual 5.7×10^{-7}), so along the series an optimised metal–donor shell responds to the metal through one scalar, the radius the tabular model already has. For 71 ligands we substitute all fifteen lanthanides into one relaxed complex, re-optimize under each Hamiltonian with one binary and one protocol, and regress the mean metal–donor distance on the Shannon radius; the slope c_L equals 1.00 when the complex contracts exactly as the ionic radii do (Figure 1c). GFN2-xTB gives $c_L = 0.405 \pm 0.145$. g-xTB [10], with the f electrons in the valence, gives 1.078 ± 0.094 and is closer to 1.00 on 70 of 71 ligands. Ln-xTB [11], an element-by-element refit of the GFN2 lanthanide parameters, recovers the mean (1.081 ± 0.214) but not a smooth series: a line explains 43 % of its metal dependence against 82 % for g-xTB, with a −0.10 Å dip at gadolinium shared by every ligand.

Fixing the contraction does not, by itself, change what a model can learn. g-xTB geometries in place of GFN2-xTB change the score by +0.005 ($p = 0.68$, eight paired seeds); rebuilding the complexes in correspondence or relaxing them in solvent makes it worse (Appendix B). Under every Hamiltonian the ligand-specific compliance does not correlate with measured selectivity ($r = 0.11, -0.02, 0.17$ on 44 ligands); what changes with the Hamiltonian is a profile shared by all ligands.

How much data the problem needs. Seventy-six extractants contribute scored pairs, so the obvious worry is that the model is starved. It is not: trained on a random fraction of each fold’s extractants, test folds untouched, the score climbs from +0.15 with eight extractants to +0.30 with forty-six and then flattens at +0.31 with all sixty-one (Figure 1d). The last third of the data is worth +0.005, inside seed noise; the two fitted extrapolations put the gain from doubling the extractant count between +0.01 and +0.04. A few dozen well-measured ligands are enough; what is missing is what separates ligands within one class (Appendix D).

Limitations and conclusion. The 3D gain is small against seed variability and rests on a single held-out test. No ceiling is identified: separations reproduce to about 0.16 on a spread of 0.22, but 70 % of scored pairs have no replicate, and within one extractant the model’s series shape is weak (median per-extractant $R^2 = 0.09$). What it predicts well is which ligands are selective and where their large separations lie. The step that mattered was asking the model the right question: the shape of a series, not its level.

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A Protocol

Code, out-of-fold predictions and every negative result: <https://github.com/mironovb/lanthanidestrain>.

Blocks. The ten binned condition descriptors that define a block are acid class, acid strength, nitrate activity, pH, diluent family, modifier class, temperature, contact time, phase ratio and metal concentration. Pairs with a measured logSF of exactly zero, which arise when both metals sit at a detection limit, are excluded. Promethium is absent, so Nd–Sm is not an adjacent pair.

Hyperparameters. CatBoost: 1,500 iterations, learning rate 0.04, depth 9, ℓ_2 regularisation 3, feature subsampling 0.3, quantile loss $\alpha = 0.6$, identical for the level and shape models. 3D encoder: width 96, three continuous-filter layers, 64 radial basis functions to 5 Å, cutoff 4 Å, heavy atoms only, dropout 0.15, head width 256, Adam at 2×10^{-3} with weight decay 10^{-4} , 60 epochs with early stopping on a validation fold, batches of 64 rows grouped by block so the contrast term always has pairs, pair-contrast weight 4 with adjacent pairs weighted $3\times$. Blend weight: nested selection over $\{0, 0.05, \dots, 1\}$ with equal weight per extractant.

Confirmation set. The 444 pairs come from lanthanide rows with computed but uncurated geometries; their identities were recorded before any 3D model was trained on the expanded population, and they were scored once with w fixed at 0.35.

Learning curve. Fractions 0.125, 0.25, 0.5, 0.75 and 1 of each fold’s training extractants, nested within a draw. The full-data run reproduces the stored champion exactly. Because the anchor is one number per extractant it cancels in every adjacent pair, so the level model does not affect the score and only the shape model’s training set matters.

B Controls

Interventions under the same protocol, development pairs; ΔR^2 relative to the best comparable system.

Intervention	ΔR^2
Simplicial encoder instead of the distance graph	0.00 ($r = 0.96$, blend weight 0.01)
Persistence descriptors as extra shape features	−0.03 to −0.09
Metal-feature weight $\times 2$ / $\times 5$ / $\times 10$	+0.00 / −0.05 / −0.08
All-pairs difference model added to the blend	0.00
Condition-transfer features	+0.004
g-xTB geometries instead of GFN2-xTB (802 complexes)	+0.005 ($p = 0.68$)
Complexes rebuilt in correspondence	−0.013
Solvent-relaxed geometries, water / octanol	−0.008 / −0.081
GFN2 / g-xTB interaction-energy features	0.00

C Ln-xTB parameters

The Ln-xTB parameters are given in the supporting information of Zhang [11] as PDF tables only. We parsed both copies (Tables S60–S61 and S62; no disagreement), replaced the $Z = 57\text{--}71$ blocks of the stock GFN2-xTB parameter file, and ran unmodified xtb 6.7.1. Single points on the 35 optimised structures shipped with the paper reproduce the published energies to better than $2 \times 10^{-8} E_h$ once the paper’s spin convention is used (the high-spin count of unpaired f electrons, reduced by one where its parity is inconsistent with the electron count). All 2,130 optimisations of the benchmark converged. A closed-shell control gives $c_L = 0.981 \pm 0.140$ with the same jagged series.

D Americium

SAFE holds 1,794 Am(III) rows on 110 ligands, 95% of them ligands with lanthanide data, giving 312 Am/Eu separation factors on 82 extractants under identical conditions. Zero-shot, with Am treated as a lanthanide at its radius, the Am/Eu separation is not predicted ($R^2 = -0.14$). Joint training with the Am rows reaches +0.47 on held-out extractants, but a control that assigns each held-out extractant the mean of the other extractants in its ligand family scores +0.47 as well, and within a family the model has no skill (+0.02); the lanthanide score is unchanged (+0.002 $\pm$ 0.020 over paired seeds). The model learns which ligand classes discriminate Am from Eu, nothing finer.